Habib Shakour

1615 Abigail Way | Ann Arbor, MI 48103
hshakour@umich.edu | (734) 386-9270 | Status: US Citizen

Education

University of Michigan - Ann Arbor **Ann Arbor, MI**
Bachelor of Science (BS in Computer Science) August 2026

- GPA: 3.6/4.0
- Course Highlights: Data Structures and Algorithms, Operating Systems, Intro to AI, Computer Security, Foundations of Computer Science, Database Management Systems, Web Systems, Computer Networking

Professional Experience

Cadence Design Systems **San Jose, CA**
Software Engineering Intern - System Verification Group (SVG) May 2025 – Aug 2025

- Designed an AI tool to configure verification IP parameters to customer devices with greater speed and accuracy
- Presented Autoconfig AI project results to Cadence executives, demonstrating business and technical value
- Collaborated globally with Indian + Brazilian teams to coordinate project responsibilities for maximum productivity
- Gained hands-on experience with EDA workflows, leveraging tools for circuit design, verification, and optimization

University of Michigan Department of Mechanical Engineering **Ann Arbor, MI**
Research Assistant June 2023 – Nov 2025

- Optimized data measuring for a machine learning algorithm through investigating fNIRS Technology
- Developed C++ code in Unreal Engine to simulate disaster relief with machine learning using VIVE VR headsets
- Demonstrated Unreal Engine simulation to PCAST, addressing questions to clarify technical aspects
- Trained incoming students on lab development tools and workflows to accelerate onboarding

Organizations

Michigan Autonomous Aerial Vehicles **Ann Arbor**
Software Team Lead Sept. 2023 – Present

- Directed a team of 4 students to create a functioning, autonomous navigation flight stack to fly any given course
- Co-developed a Computer Vision algorithm that identifies airdrop locations using Python with Open CV
- Built a functional simulation of SUAS course with drone functionality for testing C++ flight algorithms through ROS
- Implemented a comprehensive plan to maximize the productivity of 10 team members over a 12-week period

Skills

- **Programming Languages:** C++, C , Python, Java, R, MATLAB, JavaScript, SQL, HTML, CSS, System Verilog
- **Software/Applications:** Robot Operating System (ROS), LaTeX, OpenCV2, TensorFlow, PyTorch, Unreal Engine 4, Git, REACT, REST, SQL*Plus, mongoDB, AWS, gdb, Linux, UVM, Visual Studio Code, Jira

Projects

Social Media Application Prototype **Sep. 2024**

- Developed an Instagram-like application using client-side dynamic pages and a REST API in a team setting
- Managed a relational database of users' info and social media posts using SQLite to ensure data integrity
- Implemented UI features such as infinite scroll, like and comment functionality, enhancing user experience
- Ensured project integrity through end-to-end testing with Cypress and deployed project to AWS for scalability

MapReduce Framework for Distributed Processing **Nov. 2024**

- Utilized distributed processing on a cluster of computers (AWS EMR) for executing MapReduce programs
- Implemented a Manager-Worker architecture to handle job distribution, fault tolerance, and concurrency
- Achieved efficient processing by concurrently managing multiple workers that handle map and reduce tasks

Article Search Engine **Dec. 2024**

- Implemented a segmented inverted index of over 3000 articles using a hadoop MapReduce pipeline
- Ran multiple Index servers with Search server concurrently using multithreading for efficient lookup
- Built a user-friendly web interface using server-side dynamic pages for easy search and display of results